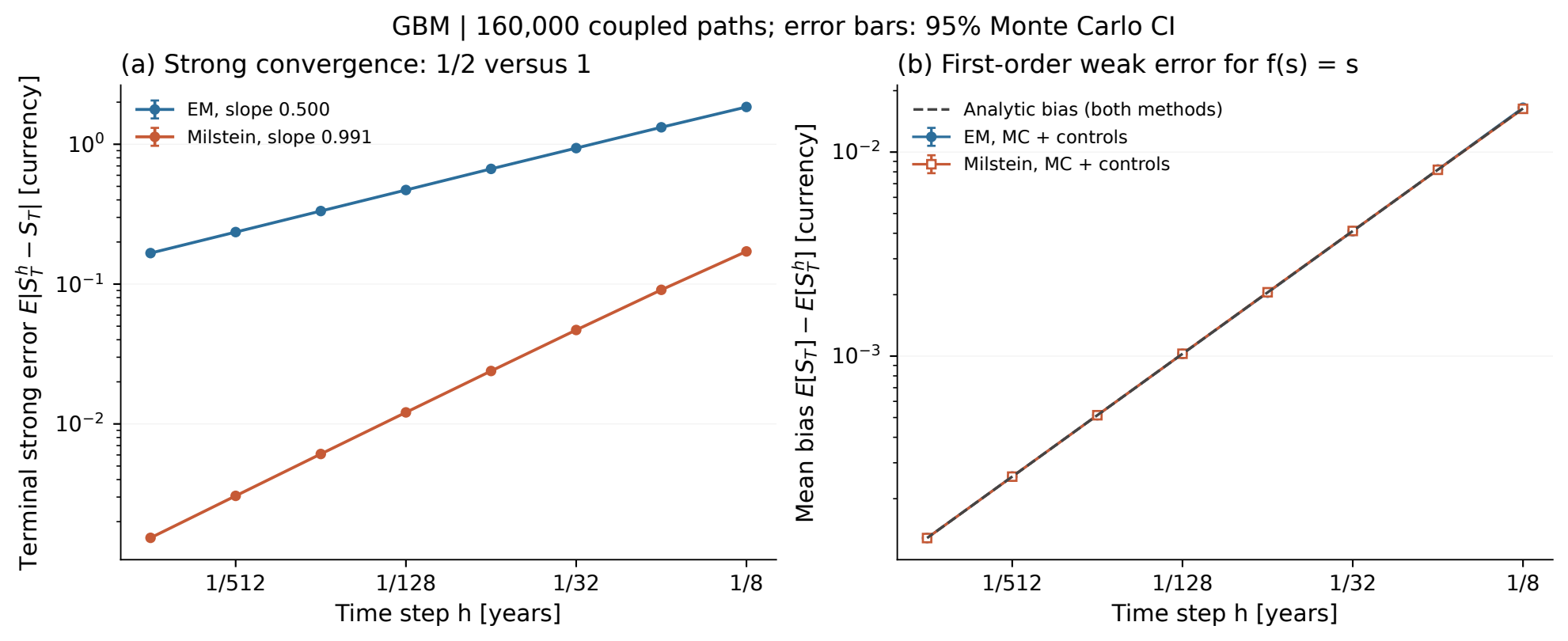
GBM | 160,000 coupled paths; error bars: 95% Monte Carlo CI
(a) Strong convergence: 1/2 versus 1
EM, slope 0.500
Milstein, slope 0.991
Terminal strong error $E|S_T^h - S_T|$ [currency]
10^0
10^{-1}
10^{-2}
1/512
1/128
1/32
1/8
Time step h [years]
(b) First-order weak error for f(s) = s
Analytic bias (both methods)
EM, MC + controls
Milstein, MC + controls
Mean bias $E[S_T] - E[S_T^h]$ [currency]
10^{-2}
10^{-3}
1/512
1/128
1/32
1/8
Time step h [years]